

Chapter 1 Measure

1.1 Construction and Extensions of Measures

1.1.1 Algebras, Semi-Algebras

Algebras

Definition 1.1.1.1 Let X be arbitrary set, then a collection $\mathcal{A}$ of subsets of X is an algebra on X if:

- $X, \emptyset \in \mathcal{A};$ (A I)

-

$$A, B \in \mathcal{A} \Rightarrow \begin{cases} A \cap B \in \mathcal{A} & \text{(A II)} \\ A \cup B \in \mathcal{A} & \text{(A III)} \\ A - B \in \mathcal{A} & \text{(A IV)} \end{cases}$$

♠ **Proposition 1.1.1.2** For algebra axiom, define

$$B \in \mathcal{A} \Rightarrow X - B \in \mathcal{A} \quad (\text{A iv})$$

Then we have

$$\{(AI), (AII), (AIII), (AIV)\} \text{ equivalent to } \{(AI), (AII), (AIII), (Aiv)\}$$

σ -Algebra

Definition 1.1.1.3 An algebra of sets $\mathcal{A}$ is called a σ -algebra if for any sequence of sets $(A_n)_{n \in \omega}$, we have

$$A_n \in \mathcal{A} \Rightarrow \cup A_n \in \mathcal{A} \quad (\text{SA I})$$

Then

Definition 1.1.1.4 ...

Generating

Definition 1.1.1.5 Let X be a set. For any family $\mathcal{F}$ of subsets of X , there exists a unique σ -algebra generated by $\mathcal{F}$, denoted by symbol

$$\sigma(\mathcal{F})$$

♠ **Proposition 1.1.1.6**

Morphisms

Borel σ -Algebra

Ring

1.1.2 Measure

Additivity, Subadditivity; Countable Additivity

Definition 1.1.2.1 Let A be a set, take the notion of power set $P(A)$, then, a **real-valued set function** μ is defined on a class of sets $\mathcal{A}$ is called **additive** if

...

Compact Classes, and Countable Additivity

Infinite, and σ -Infinite Measure

1.1.3 Lesbegue Measure

Lesbegue Measure

Lesbegue-Stieltjes Measures

1.1.4 Monotone and σ -additive classes of sets

Chapter 2 The Lebesgue Integral

2.1 Measurable Functions

Chapter 3 Operations on Measures and Functions

Chapter 4 The Space L^p and Spaces of Measures

4.1 Space L^p

Definition 4.1.0.1 Let $(X, \mathcal{A}, \mu)$ be a measure space, with nonnegative measure μ , and $p \in [1, \infty)$. Then denote

$$\mathcal{L}^p(\mu) := \{f : X \rightarrow [-\infty, \infty] \mid |f|^p \text{ is } \mu\text{-integrable}\}$$

Define μ -equivalent: The sets $\mathcal{L}^p(\mu)$ are equipped with the equivalent relation, μ -almost everywhere equivalent

$$f \sim g \Leftrightarrow \mu(\{x \in X \mid f(x) \neq g(x)\}) = 0$$

Then, denote $L^p(\mu)$ as the factor-space of $\mathcal{L}^p(\mu)$ with respect to the equivalence relation.

The norm structure is specified by

$$\| \cdot \|_p :$$

Norm $\| \cdot \|_p$

◇ **Lemma 4.1.0.2** ...

◦ **Proof** ...

□

★ **Corollary 4.1.0.3** The function $\| \cdot \|_p$ is a norm on the space $L^p(\mu)$.

◦ **Proof** ...

□

► **Theorem** ...

◦ **Proof** ...

□

4.1.1 Approximation of L^p .

4.1.2 The Weak Topology in L^p

4.2 The Hilbert Space L^2

4.3 Uniform Integrability

4.4 Convergence of Measures

Chapter 5 Integral and Derivative

5.1 Differentiability of Functions on the Real Line

Chapter 6 Borel,Baire and Souslin Sets

Chapter 7 Measures on Topological Spaces

7.1 Measures on Manifolds

Chapter 8 Weak Convergence of Measure

Chapter 9 Transformations of Measures, and Isomorphisms

Chapter 10 Conditional Measures and Conditional Expectation